# On proposals for SMC

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## 1 Feynman-Kac framework for generic SMC

We follow the notations of [Dau and Chopin, 2022]. The sequence of target distributions for $t = 0, \dots, T$, is

$$\pi_t(\mathrm{d}x) = \frac{1}{L_t}\gamma_t(x)\nu(\mathrm{d}x), \tag{1}$$

with $\nu(\mathrm{d}x)$ a base probability measure, $\gamma_t$ is a measurable non-negative function, $L_t \coloneqq \int \gamma_t(x)\nu(\mathrm{d}x)$ is the normalization constant of $\pi_t$. Let $G_t \coloneqq \gamma_t/\gamma_{t-1}$ for $t \geq 1$ and $G_0 = \gamma_0$ be the reweighting functions. Let $M_t(x_{t-1}, \mathrm{d}x_t)$ be a sequence of Markov kernels, s.t. for any $t$, $M_t$ leaves $\pi_{t-1}$ invariant.

## 2 Objective

Let $g : X \times X \to \mathbb{R}$ be some function, and for any $t \geq 0$, let

$$\begin{aligned} E(t) &\coloneqq \mathbb{E}_{X_t \sim \pi_t(\mathrm{d}x), Y_{t+1} \sim M_{t+1}(X_t, \mathrm{d}y)}[g(X_t, Y_{t+1})] \\ &= \mathbb{E}_{X_t \sim \pi_t(\mathrm{d}x)}[\mathbb{E}_{Y_{t+1} \sim M_{t+1}(X_t, \mathrm{d}y)}[g(X_t, Y_{t+1}) \mid X_t]] \end{aligned} \tag{2}$$

Given $E(t)$ monitors the quality of the proposal $M_{t+1}$, we aim to construct $M_{t+1}$ so to optimize $E(t)$.

## 3 Adaptive Waste-Free SMC

At iteration $1 \leq t \leq T$, for the particle $m$ and two consecutive particles with same ancestor $1 \leq m \leq M$, $2 \leq p \leq P$,

$$(\tilde{X}_t^{m,p-1}, \tilde{X}_t^{m,p}) \sim \pi_{t-1} \otimes M_t(\tilde{X}_t^{m,p-1}, \mathrm{d}x_t). \tag{3}$$

Let $t \geq 1$ and $\hat{\pi}_t \coloneqq \sum_{n=1}^N \delta_{X_t^n} W_t^n$, $N = MP$, where $(X_t^n)_{1 \leq n \leq N}$ are the gathered particles $\tilde{X}_t^{m,p}$ for $1 \leq p \leq P$, $1 \leq m \leq M$, then $\hat{\pi}_t$ is an estimate of $\pi_t$. Let $(M_{t+1}^\theta)_{\theta \in \Theta}$ be a collection of Markov kernels leaving $\pi_t$ invariant and let $E(t, \theta)$ be (2) with the dependency in $\theta$ made explicit. We aim to construct an estimate of $E(t, \theta)$, let it be $\hat{E}(t, \theta)$, using the samples of iteration $t$ (3) then compute

$$\hat{\theta}_{t+1}^* = \mathrm{argmax}_{\theta \in \Theta} \hat{E}(t, \theta). \tag{4}$$

We propose, for any $x \in X$, to perform importance sampling from $M_t(x, \mathrm{d}y)$ to $M_{t+1}^\theta(x, \mathrm{d}y)$ (assuming the density ratio $\mathrm{d}M_{t+1}^\theta(x, y)/\mathrm{d}M_t(x, y)$ exists) to estimate the inner expectation $\mathbb{E}_{Y_{t+1} \sim M_{t+1}^\theta(x, \mathrm{d}y)}$, and to estimate the outer expectation $\mathbb{E}_{X_t \sim \pi_t(\mathrm{d}x)}$ by the empirical reweighted mean over the particles. For the inner expectation, for all $x \in X$, $\theta \in \Theta$

$$\mathbb{E}_{Y_{t+1} \sim M_{t+1}^\theta(x, \mathrm{d}y)}[g(x, Y_{t+1})] = \mathbb{E}_{Y_t \sim M_t(x, \mathrm{d}y)}[g(x, Y_t) w_t^\theta(x, Y_t)], \tag{5}$$

with $w_{t+1}^\theta(x, y) \coloneqq \mathrm{d}M_{t+1}^\theta(x, y)/\mathrm{d}M_t(x, y)$. Replacing this expectation by the unbiased estimate obtained with one-sample, and estimating the outer expectation using the reweighted mean over the particles, we obtain Finally, an unbiased estimate $\hat{E}(t, \theta)$ is

$$\hat{E}(t, \theta) \coloneqq \frac{1}{N} \sum_{1 \leq p \leq P, 1 \leq m \leq M} W_t^{m,p-1} g(\tilde{X}_t^{m,p-1}, \tilde{X}_t^{m,p}) w_{t+1}^\theta(\tilde{X}_t^{m,p-1}, \tilde{X}_t^{m,p}). \tag{6}$$

Y: we need to do IS from $M_t(x', \mathrm{d}y)$ with $x' \sim \pi_t$ to $M_{t+1}^\theta(x, \mathrm{d}y)$ with $x \sim \pi_t$ To avoid importance sampling issue, we can consider restrictive objectives or penalised objectives to replace (**??**), taking into account the effective sample size, the variance of the weights, or replacing $\Theta$ by a neighbourhood centered around $\hat{\theta}_t$.

**MH kernel** At time $t \geq 1$, let $Q_t(x, \mathrm{d}y)$ be some proposal distribution conditionally on $x \in X$. The associated kernel is

$$M_t(x, \mathrm{d}y) = \delta_x(\mathrm{d}y) \left( \int (1 - a_t(x, y)) Q_t(x, \mathrm{d}y) \right) + a_t(x, y) Q_t(x, \mathrm{d}y), \tag{7}$$

with $a_t(x, y) = 1 \wedge \frac{\pi_{t-1}(y) Q_t(y,x)}{\pi_{t-1}(x) Q_t(x,y)}$ the Metropolis-Hasting ratio. While the ratio of MH kernels might be undefined, we can circumvent this issue by using a rao-Blackwellised estimate for $E$. Indeed, for any $t \geq 0$, we have

$$E(t) = \mathbb{E}_{X_t \sim \pi_t(\mathrm{d}x)} \left[ \mathbb{E}_{\bar{X}_{t+1} \sim Q_{t+1}(X_t, \mathrm{d}\bar{x})} [a_{t+1}(X_t, \bar{X}_{t+1}) g(X_t, \bar{X}_{t+1}) \mid X_t] \right] \tag{8}$$

using proposals $\bar{X}_{t+1} \sim Q_{t+1}(X_t, \mathrm{d}\bar{x})$ before the acceptation-rejection procedure. For any $t \geq 1$, $2 \leq p \leq P$, $1 \leq m \leq M$, let $\bar{X}_t^{m,p} \sim Q_t(\tilde{X}_t^{m,p-1}, \cdot)$ and $\bar{X}_t^{m,1} = X_{t-1}^{A_t^m}$ where $A_t^m \sim \mathrm{resample}(M, W_{t-1}^{1:N})$ be the proposed particles before acceptation-rejection step at iteration $t$. Then an estimate of $E(t, \theta)$ is given by

$$\begin{aligned} \hat{E}_t^\theta &= \frac{1}{M(P-1)} \sum_{2 \leq p \leq P, 1 \leq m \leq M}^{P} \bar{W}_t^{m,p-1} \|\tilde{X}_t^{m,p-1} - \bar{X}_t^{m,p}\|^2 a_{t+1}^\theta(\tilde{X}_t^{m,p-1}, \bar{X}_t^{m,p}) w'_{t+1}{}^\theta(\tilde{X}_t^{m,p-1}, \bar{X}_t^{m,p}), \\ w_{t+1}^{\theta,\prime}(x, y) &\coloneqq \frac{Q_{t+1}^\theta(x, y)}{Q_t(x, y)}, \; a_{t+1}^\theta(x, y) \coloneqq 1 \wedge \frac{\pi_t(y) Q_{t+1}^\theta(y, x)}{\pi_t(x) Q_{t+1}^\theta(x, y)}, \; \bar{W}_t^{m,p-1} \propto W_t^{m,p-1}, \sum_{p=2}^{P} \sum_{m=1}^{M} \bar{W}_t^{m,p-1} = 1. \end{aligned} \tag{9}$$

While for calibrating $M_1^\theta$, i.e., estimating the optimal $\theta_1$, we estimate (8) using samples from $\nu$

$$\hat{E}_0^\theta = \frac{1}{N} \sum_{n=1}^{N} \|X_0^n - \bar{X}_1^n\|^2 a_1^\theta(X_0^n, \bar{X}_1^n), \; \bar{X}_1^n \sim Q_1^\theta(X_0^n, \mathrm{d}\bar{x}), \; a_1^\theta(x, y) \coloneqq 1 \wedge \frac{\nu(y) Q_1^\theta(y, x)}{\nu(x) Q_1^\theta(x, y)}. \tag{10}$$

This is (almost) free to compute since all the involved terms are already computed during the procedure at iteration $t$, except $Q_{t+1}^\theta(\bar{X}_t^{m,p}, \tilde{X}_t^{m,p-1})$ and $Q_{t+1}^\theta(\tilde{X}_t^{m,p-1}, \bar{X}_t^{m,p})$.

**RWMH kernel: Autoregressive proposal** Assume $X = \mathbb{R}^n$, let $C \succ 0$, $\nu(\mathrm{d}x)$ some base probability measure on $X$, e.g., $\nu(\mathrm{d}x) = \mathcal{N}(0, C)$. Let $\Theta = [-1, 1]$, and let for any $t \geq 1$, $x \in X$, $\rho \in \Theta$, $Q_s^\rho(x, \mathrm{d}y) = \mathcal{N}(\mathrm{d}y \mid \rho x, (1 - \rho^2) C)$. At iteration $t \geq 1$, let us denote by $\rho_1, \rho_2, \ldots \rho_t \in \Theta$ the sequence of autoregressive parameters for the proposals $Q_1, Q_2, \ldots, Q_t$. Then for any $t \geq 1$, and any $\rho \in \Theta$

$$\begin{aligned} w_{t+1}^{\rho,\prime}(x, y) &= \frac{\mathcal{N}(\mathrm{d}y \mid \rho x, (1 - \rho^2) C)}{\mathcal{N}(\mathrm{d}y \mid \rho_t x, (1 - \rho_t^2) C)} \\ &= \left( \frac{1 - \rho^2}{1 - \rho_t^2} \right)^{-n/2} \exp\left( -\frac{1}{2} \left( (y - \rho x)^\top \left( (1 - \rho^2) C \right)^{-1} (y - \rho x) - (y - \rho_t x)^\top \left( (1 - \rho_t^2) C \right)^{-1} (y - \rho_t x) \right) \right) \end{aligned} \tag{11}$$

# References


H.-D. Dau and N. Chopin. Waste-Free Sequential Monte Carlo. *Journal of the Royal Statistical Society: Series B*, 84(1):114–148, 2022. doi: 10.1111/rssb.12475. URL https://hal.science/hal-04273259. 1

**Algorithm 1:** Adaptative Waste Free SMC with MH proposals

**input :** Integers $M, P \geq 1$, $N \leftarrow MP$, a FK model (initial distribution $\nu$, functions $G_t$, parametric MH proposals $(Q_t^\theta)_{\theta\in\Theta}$), a criterion $g$

**for** $t \leftarrow 0$ **to** $T$ **do**

  **if** $t = 0$ **then**

    **for** $n \leftarrow 1$ **to** $N$ **do**

      $X_0^n \sim \nu(\mathrm{d}x_0)$

      $\bar{X}_1^n \sim Q_1^{\theta_0}(X_0^n, \mathrm{d}\bar{x})$

      $g_1^n \leftarrow \frac{\nu(\bar{X}_1^n)}{\nu(X_0^n)}$

      $d_1^n \leftarrow g(X_0^n, \bar{X}_1^n)$

    $\theta_1 \leftarrow \mathrm{argmax}_{\theta\in\Theta} \sum_{n=1}^N d_1^n \times \left(1 \wedge g_1^n \frac{Q_1^\theta(\bar{X}_1^n, X_0^n)}{Q_1^\theta(X_0^n, \bar{X}_1^n)}\right)$

    **for** $n \leftarrow 1$ **to** $N$ **do**

      $w_0^n \leftarrow G_0(X_0^n)$

    **for** $n \leftarrow 1$ **to** $N$ **do**

      $W_0^n \leftarrow w_0^n / \sum_{n=1}^N w_0^n$

  **else**

    $A_t^{1:M} \sim \mathrm{resample}(M, W_{t-1}^{1:N})$

    **for** $m \leftarrow 1$ **to** $M$ **do**

      $\tilde{X}_t^{m,1} \leftarrow X_{t-1}^{A_t^m}$

      **for** $p \leftarrow 2$ **to** $P$ **do**

        $\bar{X}_t^{m,p} \sim Q_t^{\theta_t}(\tilde{X}_t^{m,p-1}, \mathrm{d}x_t)$

        $g_{t+1}^{m,p} \leftarrow \frac{\pi_t(\bar{X}_t^{m,p})}{\pi_t(\tilde{X}_t^{m,p-1})}$

        $d_{t+1}^{m,p} \leftarrow g(\tilde{X}_t^{m,p-1}, \bar{X}_t^{m,p})$

        $q_{t+1}^{m,p} \leftarrow Q_t^{\theta_t}(\tilde{X}_t^{m,p-1}, \bar{X}_t^{m,p})$

        $U \sim \mathcal{U}[0,1]$

        **if** $U < 1 \wedge \frac{\pi_{t-1}(\bar{X}_t^{m,p}) Q_t^{\theta_t}(\bar{X}_t^{m,p}, \tilde{X}_t^{m,p-1})}{\pi_{t-1}(\tilde{X}_t^{m,p-1}) q_{t+1}^{m,p}}$ **then**

          $\tilde{X}_t^{m,p} \leftarrow \bar{X}_t^{m,p}$

        **else**

          $\tilde{X}_t^{m,p} \leftarrow \tilde{X}_t^{m,p-1}$

    **for** $n \leftarrow 1$ **to** $N$ **do**

      $w_t^n \leftarrow G_t(X_t^n)$

    **for** $n \leftarrow 1$ **to** $N$ **do**

      $W_t^n \leftarrow w_t^n / \sum_{n=1}^N w_t^n$

    $\theta_{t+1} \leftarrow \mathrm{argmax}_{\theta\in\Theta} \sum_{2\leq p\leq P, 1\leq m\leq M} W_t^{m,p-1} d_{t+1}^{m,p} \left(1 \wedge g_{t+1}^{m,p} \frac{Q_{t+1}^\theta(\bar{X}_t^{m,p}, \tilde{X}_t^{m,p-1})}{Q_{t+1}^\theta(\tilde{X}_t^{m,p-1}, \bar{X}_t^{m,p})}\right) \times \frac{Q_{t+1}^\theta(\tilde{X}_t^{m,p-1}, \bar{X}_t^{m,p})}{q_{t+1}^{m,p}}$